



Submitted By

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CSE 311

Section: 08

Introduction

The 3M Platform is a multi-vendor e-commerce platform developed to allow different vendors to sell their products through one online platform. Customers can create accounts, browse products, add items to their cart or wishlist, place orders, make payments, and track deliveries. Vendors can manage their products and sales, while administrators can manage the overall platform. The system also includes features such as product reviews, discount coupons, returns, and customer support. The main goal of the project is to make online shopping easier and more organized.

Methodology

Database Designer Role

1. Get business requirements
2. Extract entities and their attributes
3. Draw relationships between them
4. Find cardinality of each relationship
5. Draw ER Diagram

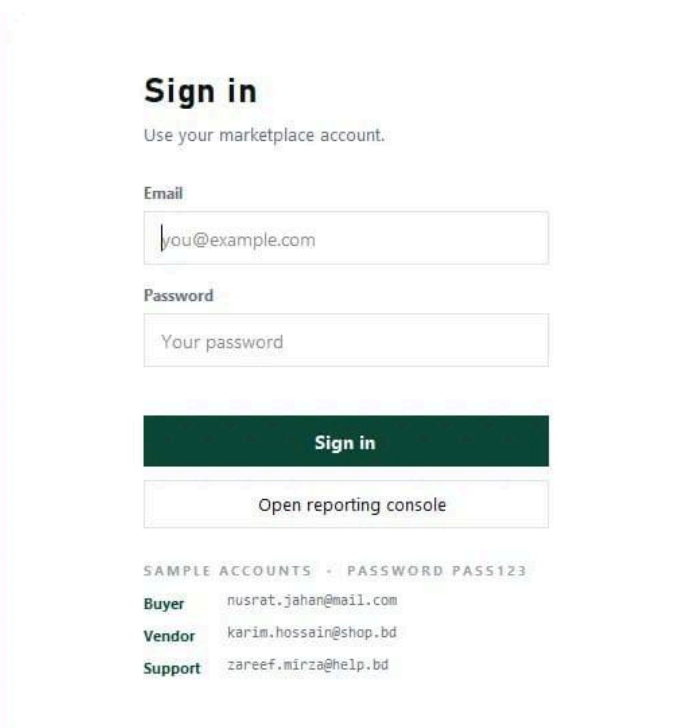
Database Administrator Role

6. Write DDI based on the ER Diagram
7. Insert some example data

Programmer

8. Write application logic so that naive users and analysts can use the system

Demonstration



M

Marketplace

Multi-vendor platform

🛒

Browse

🛒

My Cart

📦

My Orders

👤

Wishlist

👤

Profile

SU

Sohel Uddin

Buyer

Browse the catalogue

Products from every active storefront on the platform.

🔍 Search products or brands

All categories

Search

QTY

1

Add to cart

Save to wishlist

Variant id	Product	Variant	Brand	Price	Stock	Category
1356	1.5 Ton Split AC	Silver - 3 Year Warranty	Singer	Tk 68,185.65	148	Home Appliances
352	1.5 Ton Split AC	Graphite - 3 Year Warranty	Walton	Tk 56,244.88	152	Home Appliances
351	1.5 Ton Split AC	Black - 1 Year Warranty	Walton	Tk 58,358.57	173	Home Appliances
1431	1984 Paperback	Paperback - Bangla	Bloomsbury	Tk 519.57	9	English Literature
1430	1984 Paperback	Deluxe Hardcover	Bloomsbury	Tk 551.64	155	English Literature
1429	1984 Paperback	Hardcover	Bloomsbury	Tk 495.23	35	English Literature
1428	1984 Paperback	Paperback	Bloomsbury	Tk 522.00	2	English Literature
180	24 inch IPS Monitor	Black - 2 Year Warranty	Vision	Tk 15,665.16	112	TV & Monitors
179	24 inch IPS Monitor	Black - 3 Year Warranty	Vision	Tk 16,344.53	72	TV & Monitors
378	24 inch IPS Monitor	White - 1 Year Warranty	Vision	Tk 16,112.48	108	TV & Monitors
8	24 inch IPS Monitor	White - 2 Year Warranty	Sony	Tk 18,518.43	2	TV & Monitors
377	24 inch IPS Monitor	White - 2 Year Warranty	Vision	Tk 19,037.37	14	TV & Monitors
1306	24 inch IPS Monitor	White - 1 Year Warranty	Sony	Tk 15,274.39	23	TV & Monitors
7	24 inch IPS Monitor	White - 2 Year Warranty	Sony	Tk 15,565.36	164	TV & Monitors
1305	24 inch IPS Monitor	White - 2 Year Warranty	Sony	Tk 15,855.91	4	TV & Monitors

M

Marketplace

Multi-vendor platform

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Overview

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Orders

📦

My Products

⚠️

Low Stock

📢

Promotions

KH

Karim Hossain

Vendor

Sales overview

Revenue after platform commission, excluding cancelled orders.

GROSS REVENUE	ORDERS	UNITS SOLD	NEEDS RESTOCK
Tk 59 lakh	50	171	16
delivered and in flight	excluding cancelled	across all storefronts	variants below threshold

Product	Units	Revenue	Orders
TUF Gaming F15	14	Tk 2,060,586.55	7
Victus 15 Gaming	9	Tk 1,258,857.09	4
Refrigerator 324L	20	Tk 1,053,911.05	9
Vostro 3420	10	Tk 883,592.17	5
55 inch 4K Smart TV	4	Tk 274,934.35	3
1.5 Ton Split AC	2	Tk 112,489.76	2
32 inch HD LED TV	5	Tk 97,717.55	3
24 inch IPS Monitor	2	Tk 38,074.74	1
Rice Cooker 2.8L	8	Tk 32,569.02	5
Induction Cooker 2000W	5	Tk 23,657.44	2
Iron Dry 1000W	12	Tk 15,920.59	5
Electric Kettle 1.8L	8	Tk 14,146.40	5

M Marketplace
Multi-vendor platform

Browse

My Cart

My Orders

Wishlist

Profile

SU Sohel Uddin
Buyer

My orders

One row per vendor order. Select one to see how its basket was split.

Refresh

Order id	Checkout id	Store	Status
1014	640	Mamun Traders	CANCELLED
1011	639	Uddin Mart	DELIVERED
1012	639	Rasel Store	DELIVERED
1013	639	Sultana Store	DELIVERED
1009	638	Uddin Mart	DELIVERED
1010	638	Begum Mart	PACKED
1007	637	Ayesha Hub	DELIVERED
1008	637	Roy Enterprise	DELIVERED

How this basket was split

ONE CHECKOUT
#639
Tk 11,663.33
BKASH · 2026-04-19 1...

Uddin Mart

Order #1011 · 2 items

Tk 3,708.11

DELIVERED

Rasel Store

Order #1012 · 2 items

Tk 4,577.88

DELIVERED

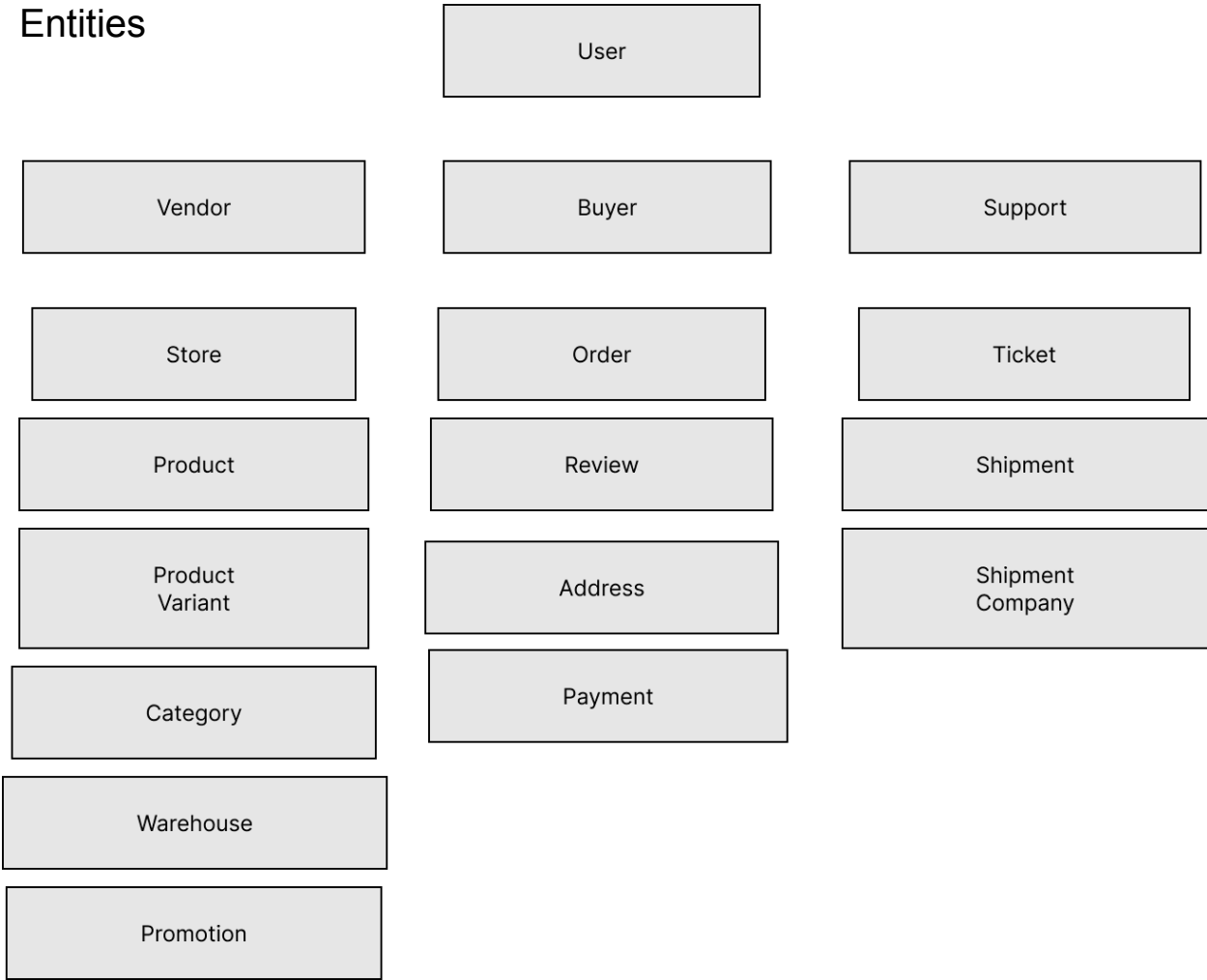
Sultana Store

Order #1013 · 2 items

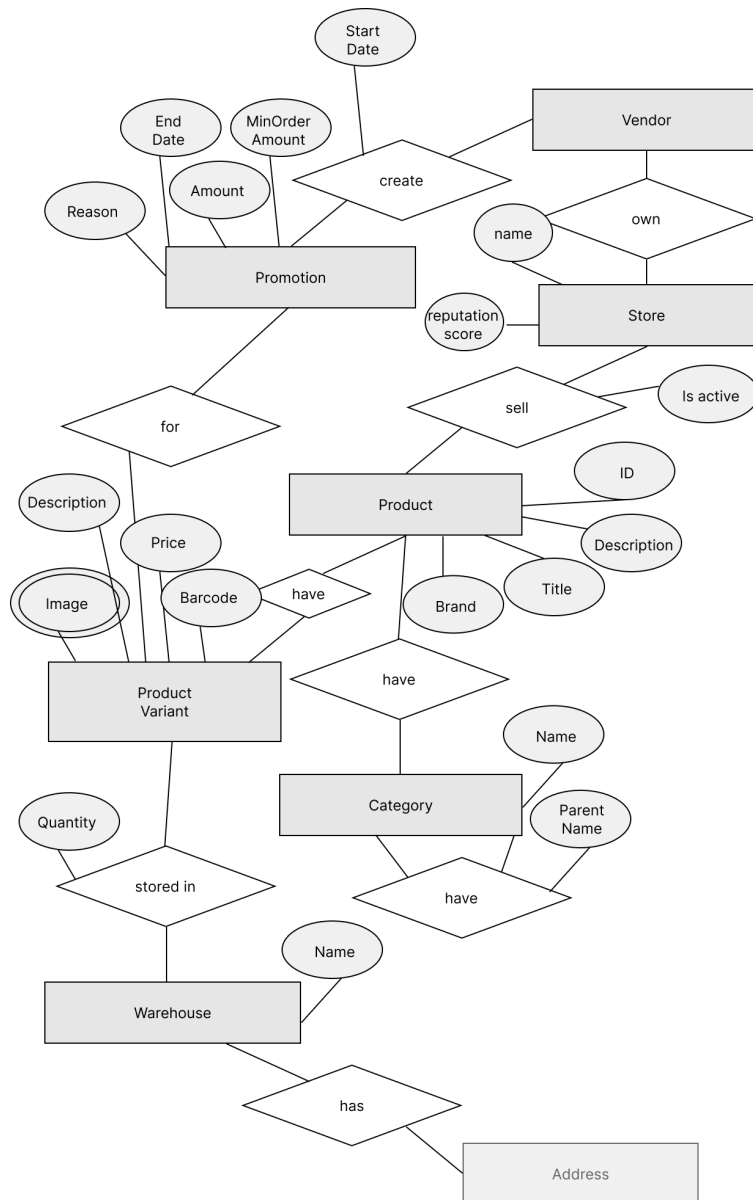
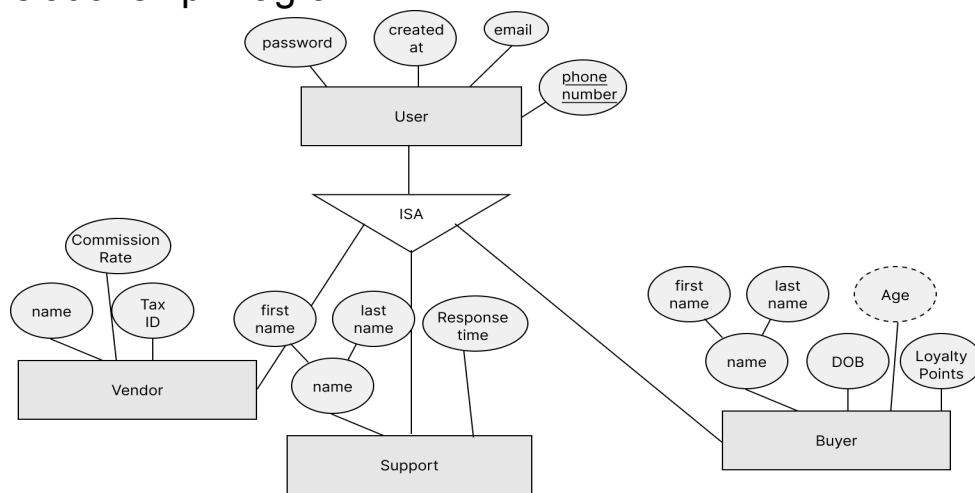
Tk 3,377.34

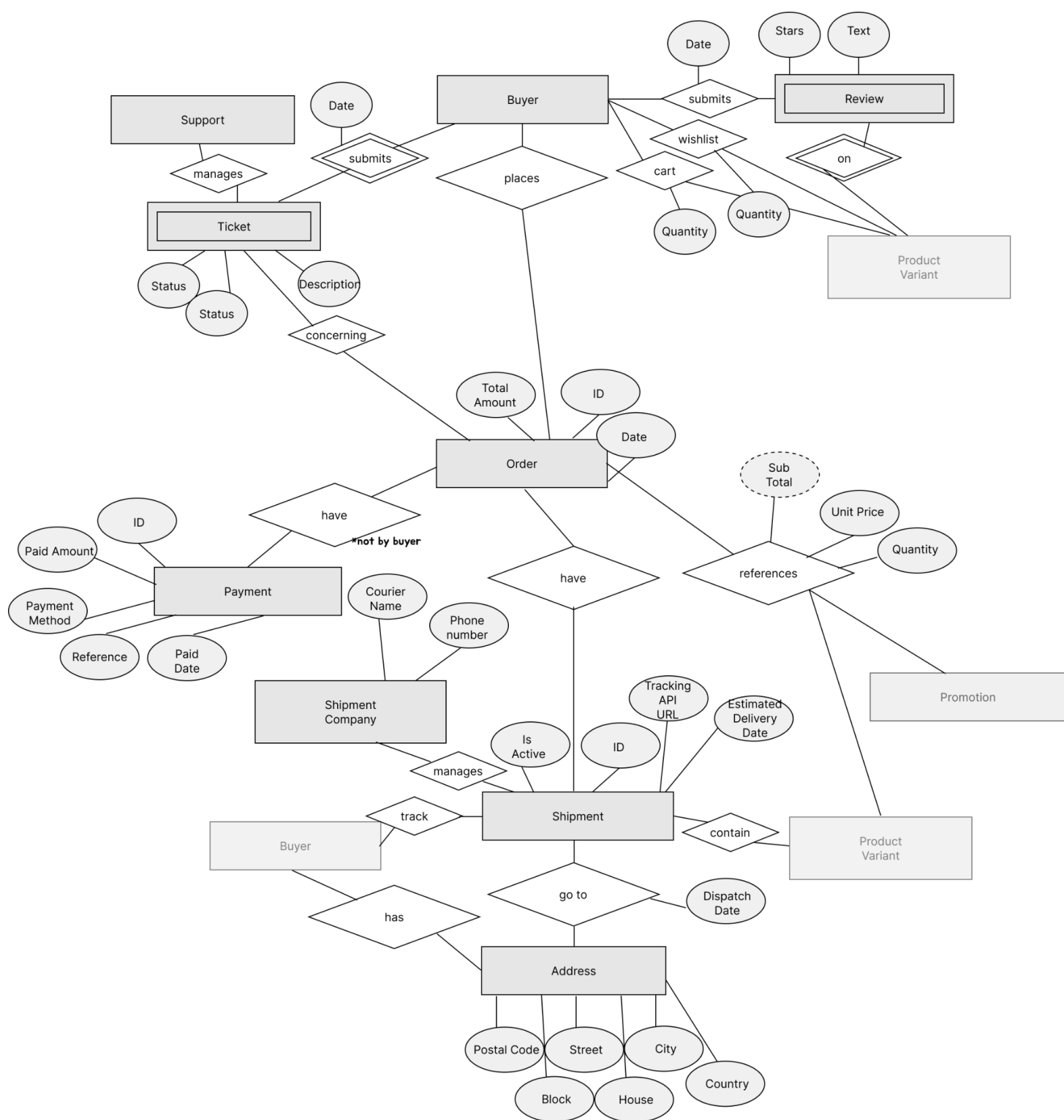
DELIVERED

Entities



Entity Relationship Diagram





limitation

We did not have image handling, authentication and security features were kept simple and do not include advanced methods such as two-factor authentication. The payment system is also faked, and would require third party integration.

Conclusion

In conclusion, developing the 3M Platform helped us learn how to design and build a relational database based on a given real-world situation. We learned about cardinality and how relationships between tables work, as well as how to normalize a database up to 3NF to reduce unnecessary data duplication. We also learned common ways of joining tables based on their relationships. Most importantly, the project improved our ability to analyze a situation and create a suitable RDBMS structure for it. Overall, this project gave us practical experience with database design and SQL.